Frank Fan

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EDUCATION

PhD, Computer Science | University of Waterloo, Waterloo, ON — Sep 2026 – Aug 2030 (expected)
Advisor: Prof. Gladimir Baranoski. Focus: physically based rendering, spectral light transport.

Master of Mathematics (MMath), Computer Science | University of Waterloo, Waterloo, ON — Sep 2024 – Apr 2026
GPA 4.0/4.0. Thesis: simulation and visualization of chlorophyll fluorescence, implemented in C++ on pbrt-v4.

Bachelor of Computer Science (BCS) | University of Waterloo, Waterloo, ON — Sep 2019 – Apr 2024
GPA 3.8/4.0. University of Waterloo Entrance Scholarship. Coursework: Algorithms, Computer Graphics, Numerical Computation.

WORK EXPERIENCE

Software Developer Intern | Rocscience Inc., Toronto, ON — Jan 2023 – Apr 2023

- As the sole developer, built the 3D Barrier Design and Analysis feature of RocFall3, a commercial rockfall simulator; the feature still ships today and is featured on the product's homepage.
- Developed C# algorithms computing the trajectory height, impact angle, velocity, and kinetic energy of thousands of simulated rock paths hitting a barrier, so engineers can size and place rockfall barriers.
- Rendered impact distributions along the barrier as interactive 3D histograms and heatmaps using OpenGL and vertex shaders through the Eyeshot .NET framework.

Software Developer Intern | Tactic Studios, Toronto, ON — May 2022 – Aug 2022

- Built a Java CI pipeline on the Perforce API for an in-house game engine, automating branch syncing and merging, remote builds, and asset baking.

Software Developer Intern | Rocscience Inc., Toronto, ON — Sep 2021 – Dec 2021

- Rearchitected C++ spatial partitioning for polygon meshes of about one million triangles, replacing a 3D uniform grid with an adaptive octree to accelerate point-mass-to-mesh collision detection in rockfall simulation.

Software Developer Intern | PSI Technologies Inc., Saskatoon, SK — Jan 2021 – Apr 2021

- Built a C# .NET WinForms tool that slices 3D models of millions of triangles along any plane, in geotechnical software used to design and construct roads.
- Implemented the real-time mesh clipping algorithm that regenerates the cross section live as the user moves the plane.

RESEARCH AND TEACHING

Graduate Researcher | University of Waterloo, Waterloo, ON — Sep 2024 – Present

- Extended pbrt-v4, a large open-source C++ renderer, with a new participating-medium class that simulates fluorescence across the light spectrum, integrating with its existing spectral multiple importance sampling.
- Implemented physically based models driven by measured data: absorption derived from measured absorbance spectra, Rayleigh scattering, and fluorescent re-emission at longer wavelengths.
- Produced, to our knowledge, the first renderings of fluorescence under combined ultraviolet and visible light (MMath thesis); co-authored a paper at the Eurographics Symposium on Rendering (EGSR) 2026.

Undergraduate Research Assistant | University of Waterloo, Waterloo, ON — Sep 2023 – Apr 2024

- Compared stochastic, spectral, and rasterization-based techniques for rendering sparkles on visual fidelity and computational cost; published the results as a University of Waterloo technical report.

Teaching Assistant | University of Waterloo, Waterloo, ON — Sep 2024 – Present

- TA for six course offerings, including three terms of CS488 Computer Graphics; as Lead TA, coordinated 2 TAs supporting about 60 students and 250 assignments.
- Explain ray tracing, rasterization, and shader debugging to students in weekly office hours and on Piazza.

PROJECTS

Recursive Ray Tracer | C++, OpenGL — 2022

- Built a recursive ray tracer supporting reflection, refraction, glossy reflection and refraction, texture mapping, bump mapping, and stochastic anti-aliasing.

TECHNICAL SKILLS

Languages: C++, C#, Python, C, Java, SQL, JavaScript, HTML/CSS, GLSL, Bash
Concepts: Data structures and algorithms, octrees, BVH, computational geometry, collision detection, OpenGL, ray tracing
Tools: Git, GitHub, Perforce, Linux, .NET, ImGui, CI/CD, Cloudflare, GitHub Pages